MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 7621

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1

Mounting position: Any

Weight: 2 ounces (56.7 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ip	egy	ib
Unit:	Vac	kv	kv	Vdc	Aac	v	a
Maximum:	6.8	8 (Note 1)	8 (Note 2)	---	2	--- (Note 1)	90
Minimum:	5.8	---	5% epy	300	---	175	---
TEST CONDITIONS:	5.8	8	---	---	---	130	---

ABSOLUTE RATINGS:

Parameter:	Ib	tk	Pb	TA	dik	prr	Cooling	tj	Ecc
Unit:	mAdc	sec	---	°C	a/µs	---	---	µs	Vdc
Maximum:	100	---	2.7×10^9	+125	1,000	---	Note 4	0.010	---
Minimum:	---	90	---	-65	---	---	---	---	0
TEST CONDITIONS:	---	90	---	Ambient	---	4,000	---	---	0

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3267	Instantaneous starting	5,6	Ef = 6.8 Vac	0.65	II	---	---	---	---
3246	Operation	5	t = 300	0.65	II	---	---	---	---
3241	Heater current	-	Ef = 6.3 Vac	0.65	II	If	2.2	3.5	Aac
3251	Pulse emission (Method A)	7	ik = 90a(min); prr = 60 ±10%; tp = 5.0 µs ±10%; tr = 0.5 µs(max)	0.65	II	egk	---	200	v
3247	DC anode voltage for conduction	-		0.65	II	Ebb	---	200	Vdc
	Quality conformance inspection, part 2								
3256	Anode delay time	-	Operation; t = 30	---	---	tad	---	0.4	µs
3256	Anode delay time drift	8	Anode delay time	---	---	∆tad	---	0.15	µs
3261	Time jitter	-	Operation	---	---	tj	---	0.01	µs
	Quality conformance inspection, part 3								
----	Life test	5,9	t = 200 hours	---	---	---	---	---	---
----	Life-test end points:								
3246	Operation	-	egy = 140v (max)	---	---	---	---	---	---
3247	DC anode voltage for conduction	-		---	---	Ebb	---	250	Vdc
3256	Anode delay time	-	egy = 140v (max)	---	---	tad	---	0.5	µs
3246	Operation at elevated ambient temperature	5,11, 12	t = 5 hours; TA = 90°C; Ef = 6.8 Vac	---	---	egy	---	175	v
1031	Variable-frequency vibration	10,12	No voltages applied	---	---	---	---	---	---
1042	Shock, specified pulse	12	Test condition D	---	---	---	---	---	---
----	Shock-test end points:								
3246	Operation	-		---	---	egy	---	175	v
3247	DC anode voltage for conduction	-		---	---	Ebb	---	250	v
3261	Time jitter	-		---	---	tj	---	0.01	µs

NOTES:

1. Instantaneous starting permissible. The maximum permissible egy is 8.0 kv and shall not be attained in less than 0.04 second.
2. In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 2.5 kv during the first 25 µs following the anode pulse.
3. The driver pulse, measured at the tube socket with thyratron grid disconnected shall have the following characteristics: egy = 175 v(min), tr = 0.25 µs (max), grid pulse duration tp = 1.0 µs (min). Impedance of drive circuit = 1,200Ω (max).

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4. Forced-air cooling directed on the anode or envelope is permissible, depending upon operating conditions.

5. The circuit constants shall be chosen under resonant charging conditions so that epy = 8.0 kv; ib = 90a (min); dik/dt = 1,000a/µs (min); tp = 0.12 ±0.012 µs; prr = 4,000 (min). Grid pulse as measured at tube socket with thyratron grid disconnected shall have the following characteristics: tr = 0.25 µs (max), tp = 1.0 µs (max), internal impedance of driver: 1,200 ohms (min).

6. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 8.0 kv (min) within 0.03 second. Any tube failing to start within three attempts shall be considered a failure.

7. The voltage between grid and cathode shall be measured not more than 2.5 µs after the beginning of the current pulse.

8. During the interval between 30 and 90 seconds of the anode delay time test, the (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.

9. During every 50 ±8-hour period, the life test shall be shut off for 60 minutes (min) and then tested for life-test end points.

10. There shall be no pronounced resonance in the range specified in method 1031.

11. The tube shall operate satisfactorily at the ambient temperature specified herein at operation test conditions for a total of 5 consecutive hours. This test shall be performed on a minimum of four tubes on an annual basis.

(E) 12. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

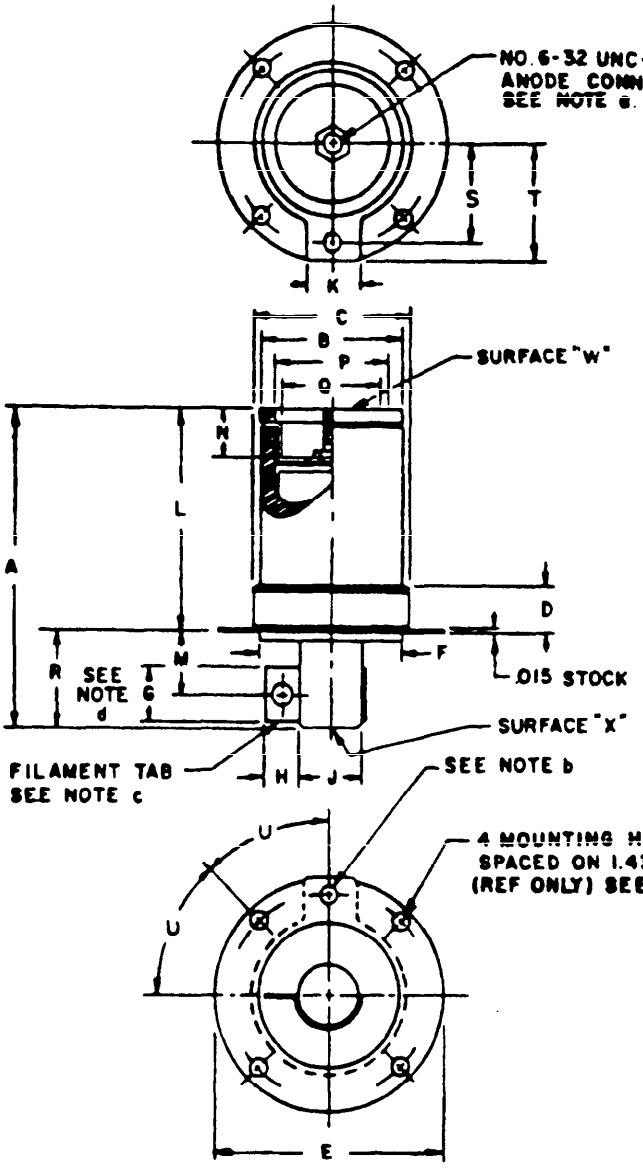
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3089)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		2.355 (59.82)
B	.985 (25.02) DIA	1.015 (25.78) DIA
C		1.150 (29.21) DIA
D	.295 (7.49)	.340 (8.64)
E	1.579 (40.11) DIA	1.639 (41.63) DIA
F	.985 (25.02) DIA	1.015 (25.78) DIA
G		.500 (12.70)
H		.438 (11.13)
J		.469 (11.91) DIA
K	.250 (6.35)	.500 (12.70)
L	1.500 (38.10)	1.600 (40.64)
M	.335 (8.51)	.585 (14.86)
N	.250 (6.35)	.360 (9.14)
P	.785 (19.94)	.815 (20.70)
Q	.650 (16.51) DIA	.710 (18.03) DIA
R		.700 (17.78)
Reference dimensions		
S	.670 (17.02)	
T	.795 (20.19)	
U	45°	

NOTES:
a. Anode connector shall not extend above surface "W".
b. All mounting holes, including grid and filament tab holes, shall clear no. 4 screws, (.120 dia holes).
c. Wrap of filament tab on protector cap optional, right or left side.
d. Tab edge shall not extend beyond surface "X".
e. Grid and filament tabs shall not interfere with cathode flange mounting holes.

FIGURE 1. Outline drawing of electron tube type 7621.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

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